

Homework 1

Chapter 1: The Real Numbers

15 problems: 5 easy, 5 medium, and 5 challenging.

Instructions. Give proofs or counterexamples as requested. You may use Chapter 1 and earlier problems, except a conclusion you are asked to prove. Do not use limit theorems, continuity, or calculus. Write your solutions on separate paper.

Notation. $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{N}^* = \{1, 2, \dots\}$, and $\mathcal{R}$ is the set of Dedekind cuts in $\mathbb{Q}$, ordered by inclusion. For a cut, $X^c = \mathbb{Q} \setminus X$.

Part A. Easy

Problem 1. Equivalence classes and well-defined operations

On $\mathbb{Z}$, define $a \sim b$ if $a^2 = b^2$, and write $[a]$ for the class of a .

- (a) Prove that $\sim$ is an equivalence relation and describe every class.
- (b) Decide whether each rule $[a] \oplus [b] = [a + b]$ and $[a] \odot [b] = [ab]$ defines a well-defined operation on $\mathbb{Z}/\sim$. Prove your answer or give a counterexample.

Problem 2. A partial order and its extremal elements

Let $D = \{1, 2, 3, 4, 6, 12\}$, ordered by divisibility: $a \leq b$ means $a \mid b$.

- (a) Verify that this is a partial order, but not a total order.
- (b) For $A = \{4, 6\}$, find all upper and lower bounds in D , all maximal and minimal elements in A , and $\sup_D A$ and $\inf_D A$. Does A have a greatest or a least element?

Problem 3. Absolute value in an ordered field

Let F be an ordered field and define $|x| = \max\{x, -x\}$. Using the ordered-field axioms, prove for all $x, y \in F$ that

$$|xy| = |x||y|, \quad |x + y| \leq |x| + |y|, \quad ||x| - |y|| \leq |x - y|.$$

Problem 4. Suprema, infima, and attainment

- (a) Find the supremum and infimum in $\mathbb{R}$ of each set

$$A = \left\{ \frac{n}{n+1} : n \in \mathbb{N}^* \right\}, \quad B = \{q \in \mathbb{Q} : -2 < q \leq 3\}.$$

Decide which are attained. Justify the bounds without using limits.

- (b) Determine all real upper and lower bounds of $\emptyset$. Explain why $\sup_{\mathbb{R}} \emptyset$ and $\inf_{\mathbb{R}} \emptyset$ do not exist as real numbers.

Problem 5. Integer parts

For $x \in \mathbb{R}$, let $\lfloor x \rfloor$ be the unique integer m with $m \leq x < m + 1$.

- (a) Prove

$$\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1.$$

Give examples in which the lower bound, respectively the upper bound, is attained.

- (b) Express $\lfloor -x \rfloor$ in terms of $\lfloor x \rfloor$, distinguishing $x \in \mathbb{Z}$ from $x \notin \mathbb{Z}$. In particular, compute $\lfloor -7/3 \rfloor$.

Part B. Medium**Problem 6. Arithmetic of suprema and infima**

Let $A, B \subseteq \mathbb{R}$ be nonempty bounded sets.

- (a) Prove that $s = \sup A$ if and only if s is an upper bound of A and, for every $\varepsilon > 0$, some $a \in A$ satisfies $s - \varepsilon < a$.
- (b) Define $A + B = \{a + b : a \in A, b \in B\}$ and $A - B = \{a - b : a \in A, b \in B\}$. Prove

$$\sup(A + B) = \sup A + \sup B, \quad \inf(A - B) = \inf A - \sup B.$$

Problem 7. Rational and irrational numbers in every interval

Let $a < b$ be real numbers.

- (a) Prove that, for every $n \in \mathbb{N}^*$ satisfying $n(b - a) > 1$, there is an integer m with $a < m/n < b$. Deduce the density of $\mathbb{Q}$ in $\mathbb{R}$.
- (b) Using only part (a) and the existence and irrationality of $\sqrt{2}$, prove that (a, b) contains infinitely many rational numbers and infinitely many irrational numbers. Do not quote the density of the irrationals.

Problem 8. Why negating a cut needs care

For $r \in \mathbb{Q}$, write $X_r = \{q \in \mathbb{Q} : q < r\}$. Let $X \subseteq \mathbb{Q}$ be a Dedekind cut and put $X^c = \mathbb{Q} \setminus X$.

- (a) Compute $\{-u : u \in X_r^c\}$ and explain why it is not a cut.
- (b) Define

$$Y = \{q \in \mathbb{Q} : q < -u \text{ for some } u \in X^c\}.$$

Prove directly that Y is a cut and that $X + Y = X_0$, where $X + Y = \{x + y : x \in X, y \in Y\}$. You may use the small-gap Lemma 1.5.2, but not the already proved additive-inverse law for cuts.

Problem 9. Nested intervals and their hypotheses

Let $I_n = [a_n, b_n] \subseteq \mathbb{R}$ be nonempty closed bounded intervals with $I_{n+1} \subseteq I_n$ for every $n \geq 1$.

- (a) With $\alpha = \sup\{a_n : n \geq 1\}$ and $\beta = \inf\{b_n : n \geq 1\}$, prove

$$\alpha \leq \beta, \quad \bigcap_{n \geq 1} I_n = [\alpha, \beta].$$

- (b) Show that the intersection is a singleton if $b_n - a_n \leq 1/n$ for every $n \geq 1$.
- (c) Give a nested family of nonempty open bounded intervals with empty intersection, and a nested family of nonempty closed unbounded intervals with empty intersection. Verify both examples.

Problem 10. A non-Archimedean ordered field

In $\mathbb{Q}(T)$, use the order from Section 1.4: a nonzero rational function $P(T)/Q(T)$ is positive when its numerator and denominator have leading coefficients of the same sign.

- (a) Prove that $0 < 1/T < r$ for every rational $r > 0$. Deduce that the canonical copy of $\mathbb{Q}$ is not order dense in $\mathbb{Q}(T)$.
- (b) Prove that $\mathbb{N}$ is bounded above in this field but has no supremum there.
- (c) Show that there is no order-preserving unital field homomorphism from $\mathbb{Q}(T)$ to $\mathbb{R}$.

Part C. Challenging

These problems develop further consequences of Chapter 1.

Problem 11. Constructing positive roots from completeness

Fix $a > 0$ and an integer $n \geq 2$.

- Show that $E = \{t \in \mathbb{R} : t \geq 0, t^n \leq a\}$ is nonempty and bounded above, and that $x = \sup E$ is positive.
- Prove that $x^n = a$ and that no other positive real number has n th power a . Do not use continuity or an intermediate value theorem.

Hint. Use the finite identity

$$u^n - v^n = (u - v) \sum_{j=0}^{n-1} u^{n-1-j} v^j$$

to control a small change in the n th power.

Problem 12. Rigidity of real arithmetic

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$f(x + y) = f(x) + f(y), \quad f(xy) = f(x)f(y) \quad (x, y \in \mathbb{R}).$$

Prove that either $f(x) = 0$ for every x , or $f(x) = x$ for every x . Do not assume that f preserves 1, is injective, preserves order, or is continuous. You may use Problem 11.

Problem 13. Infima in the cut construction

Let $\mathcal{A} \subseteq \mathcal{R}$ be a nonempty family of Dedekind cuts, bounded below in the inclusion order. Define

$$D = \bigcap_{X \in \mathcal{A}} X, \quad L = \{q \in \mathbb{Q} : q < r \text{ for some } r \in D\}.$$

- Working directly with cuts, prove that L is a cut and $L = \inf_{\mathcal{R}} \mathcal{A}$.
- For $\mathcal{A} = \{X_{1/n} : n \in \mathbb{N}^*\}$, compute D and L . Explain precisely why simply taking the intersection does not always give the infimum in $\mathcal{R}$.

Problem 14. Rational numbers and repeating decimals

Let $x \in [0, 1)$ have canonical decimal expansion $0.d_1d_2\cdots$ (not eventually 9). Call the expansion *eventually periodic* if there exist integers $N \geq 0$ and $k \geq 1$ such that $d_{j+k} = d_j$ for all $j > N$.

- If $x \in \mathbb{Q}$, prove that its canonical decimal expansion is eventually periodic. A terminating expansion counts as periodic with a tail of zeros.
- Prove the converse, using finite geometric sums and the supremum definition of a decimal, rather than an infinite-series formula.

Hint for (a). Track the remainders in long division by the denominator.

Problem 15. A countable dense set and an uncountable interval

A set is *countable* if it is finite or its elements can be listed in a sequence; repetitions in a list are allowed.

- Prove that $\mathbb{Q}$ is countable by listing pairs $(p, q) \in \mathbb{Z} \times \mathbb{N}^*$ in order of $|p| + q$.
- Given any sequence $x_1, x_2, \dots$ in $[0, 1]$, construct nonempty closed intervals $I_0 = [0, 1] \supseteq I_1 \supseteq \cdots$ such that I_n has length 3^{-n} and $x_n \notin I_n$ for every $n \geq 1$. Deduce that $[0, 1]$ is not countable.

Hint for (b). At each step, compare the left and right closed thirds of the preceding interval.